



Performance of transformer-based machine learning methods for UAV time series prediction

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Motivation

Problem: The acquisition of data in drone-based (UAV) geophysical electromagnetic (EM) surveys can be compromised by issues such as memory errors or equipment failure, leading to incomplete datasets.

Objective: The master's thesis aims to evaluate the effectiveness of transformer-based machine learning architectures, leveraging their attention mechanism, for full times series prediction. A trained network can be used to replace faulty time series and render the datasets viable for subsequent analysis and interpretation.

Training Data

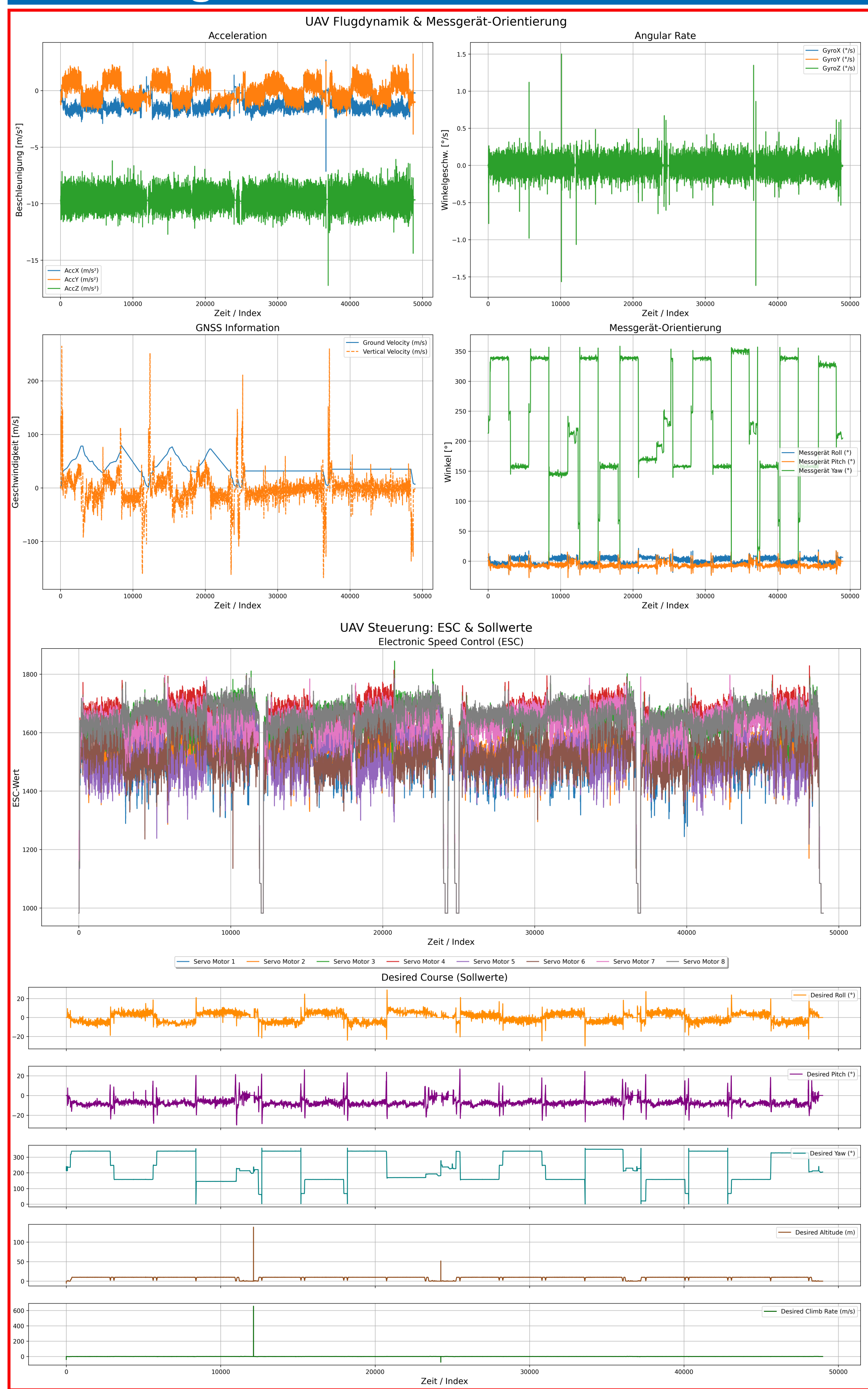


Fig. 2: Segment from the training data (measured by flight controller (Fig. 1A)). Shown is the UAV's flight dynamics and orientation (top), and corresponding control signals (bottom).

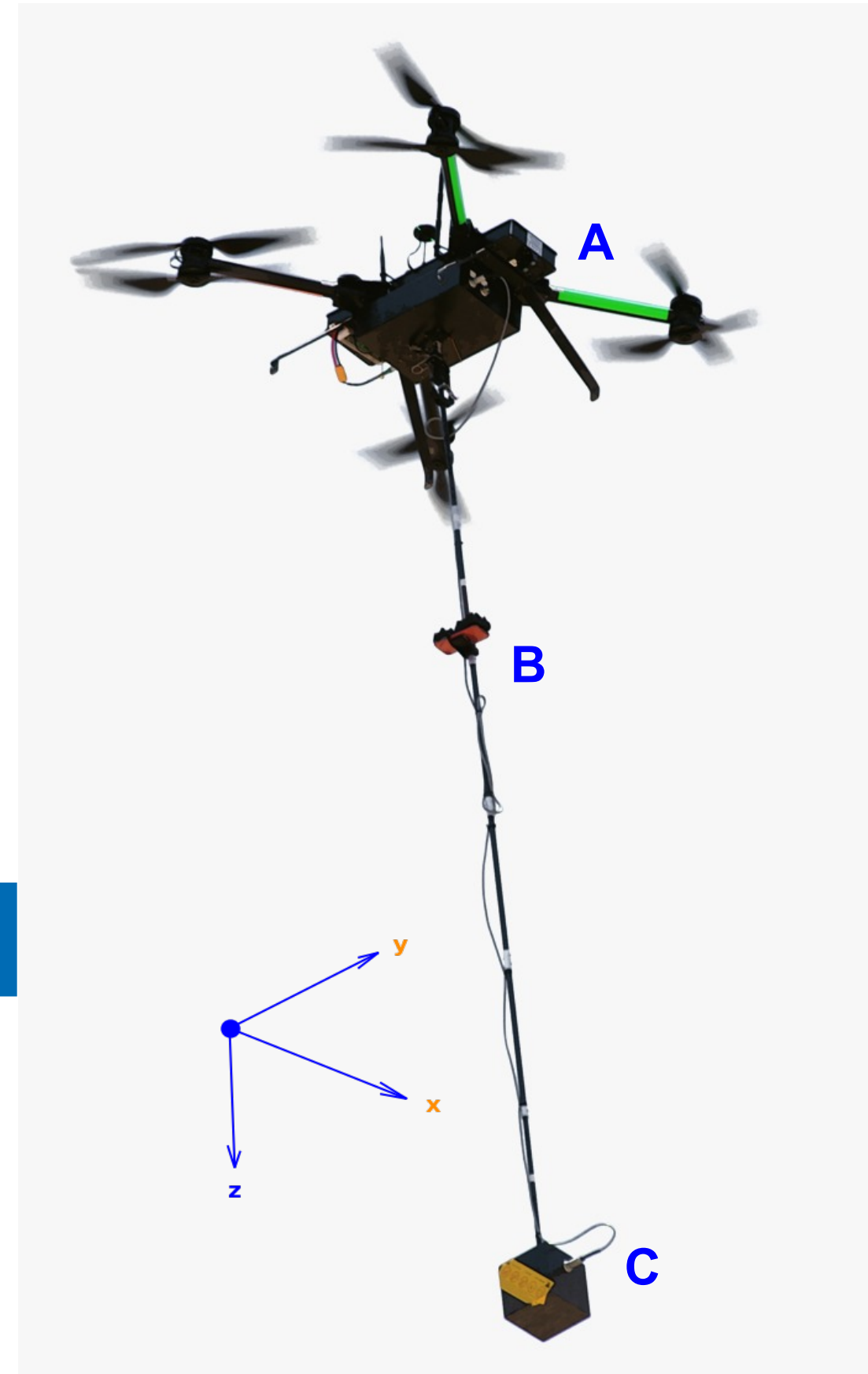


Fig. 1: UAV (A) with inertial sensor (B) and magnetometer (C) (modified after [2]).

Transformer Overview

Model Configuration:

- **Sequence Logic:** 50 input steps → 5 forecast steps
- **Hyperparameters:** Embed Dim: 128 | Heads: 4 | FFN Dim: 512 | Blocks: 2
- **Training:** Adam ($2 \cdot 10^{-4}$) | Cosine Annealing | 10-Epoch Warmup
- **Regularization:** Dropout: 0.1 | Weight Decay: $5 \cdot 10^{-4}$

Technical Implementations:

- **Geometric Encoding:** Yaw angles are projected into Sine/Cosine pairs to ensure a continuous loss landscape and prevent 360° "wrap-around" issues.
- **Hybrid Loss Function:** A combination of Huber Loss ($\beta = 2.0$) for robust regression and a custom Unit Circle Loss ($\lambda = 1.5$).
- **Constraint Enforcement:** The Unit Circle penalty explicitly enforces the trigonometric identity $\sin^2 + \cos^2 = 1$, ensuring physically plausible orientation vectors.
- **Architecture Advantage:** By leveraging 50 steps of temporal context, the Transformer effectively captures high-frequency oscillations that traditional FFNs fail to track.

Transformer Prediction

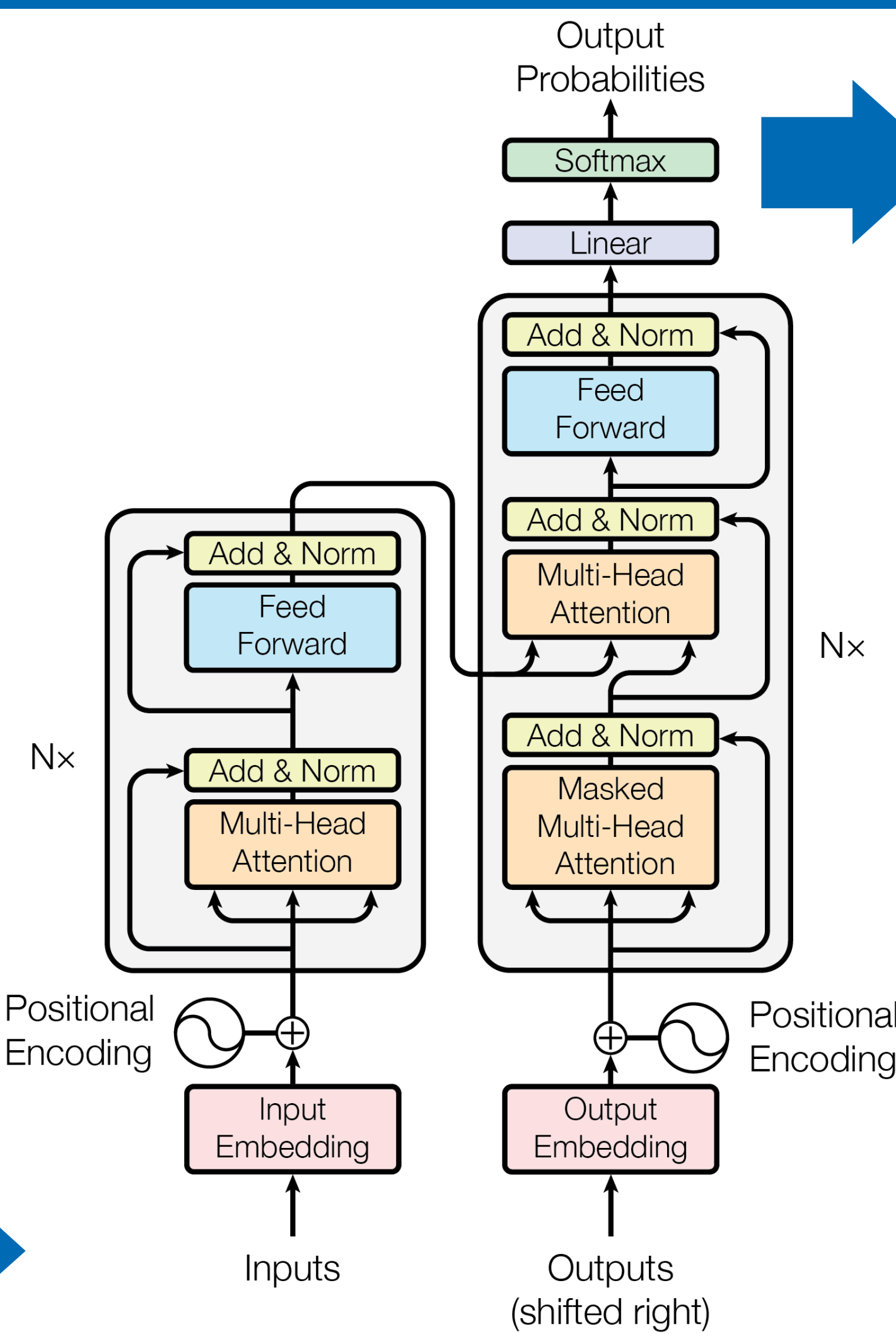


Fig. 3: The Transformer architecture [1] featuring an encoder stack (left) and a decoder stack (right).

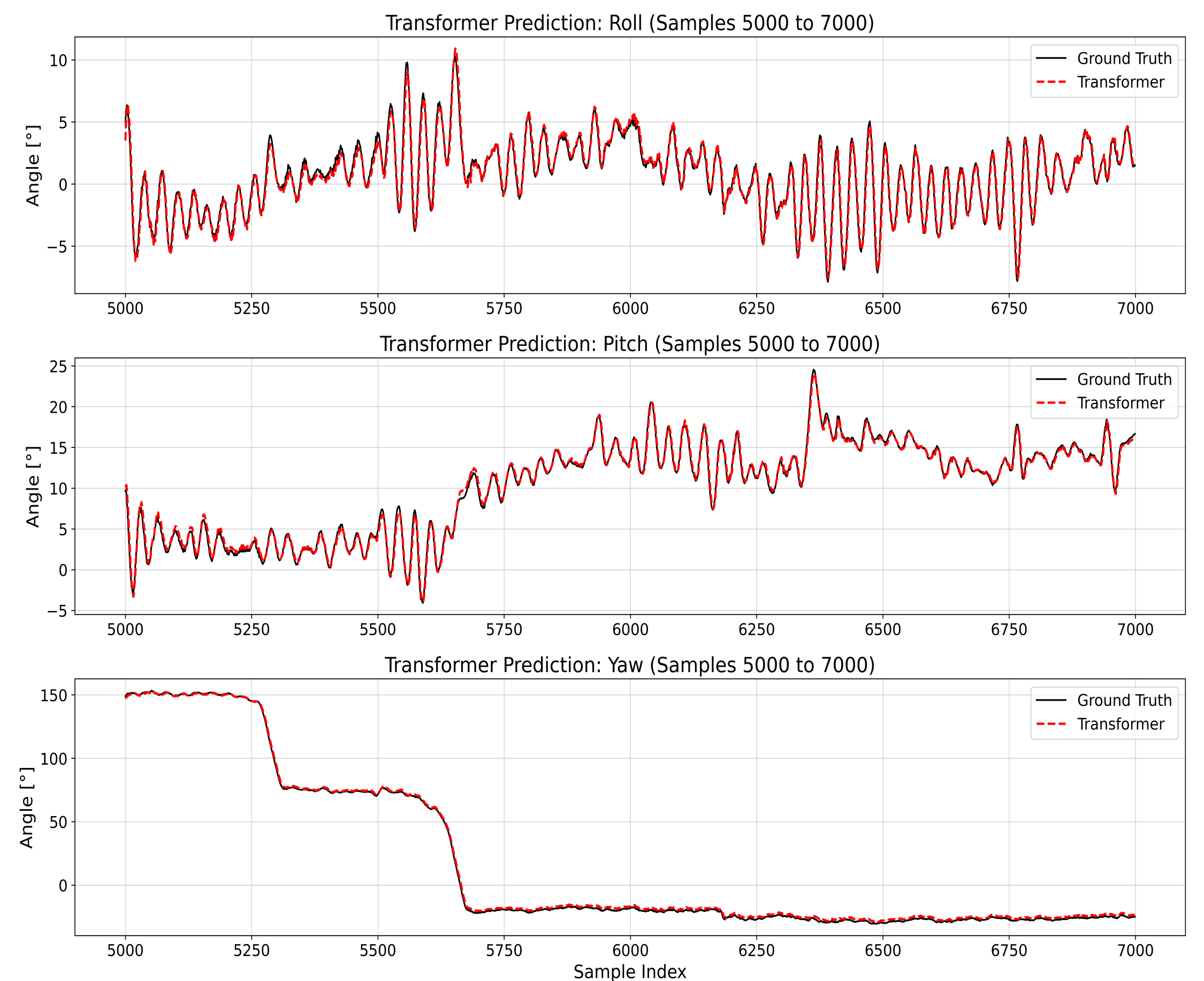


Fig. 4: Performance of the Transformer model on a test data segment. The plots compare the model's predictions (red, dashed line) against the true values (black, solid line) for the Roll, Pitch, and Yaw orientation angles (measured by Fig. 1B).

Residuals

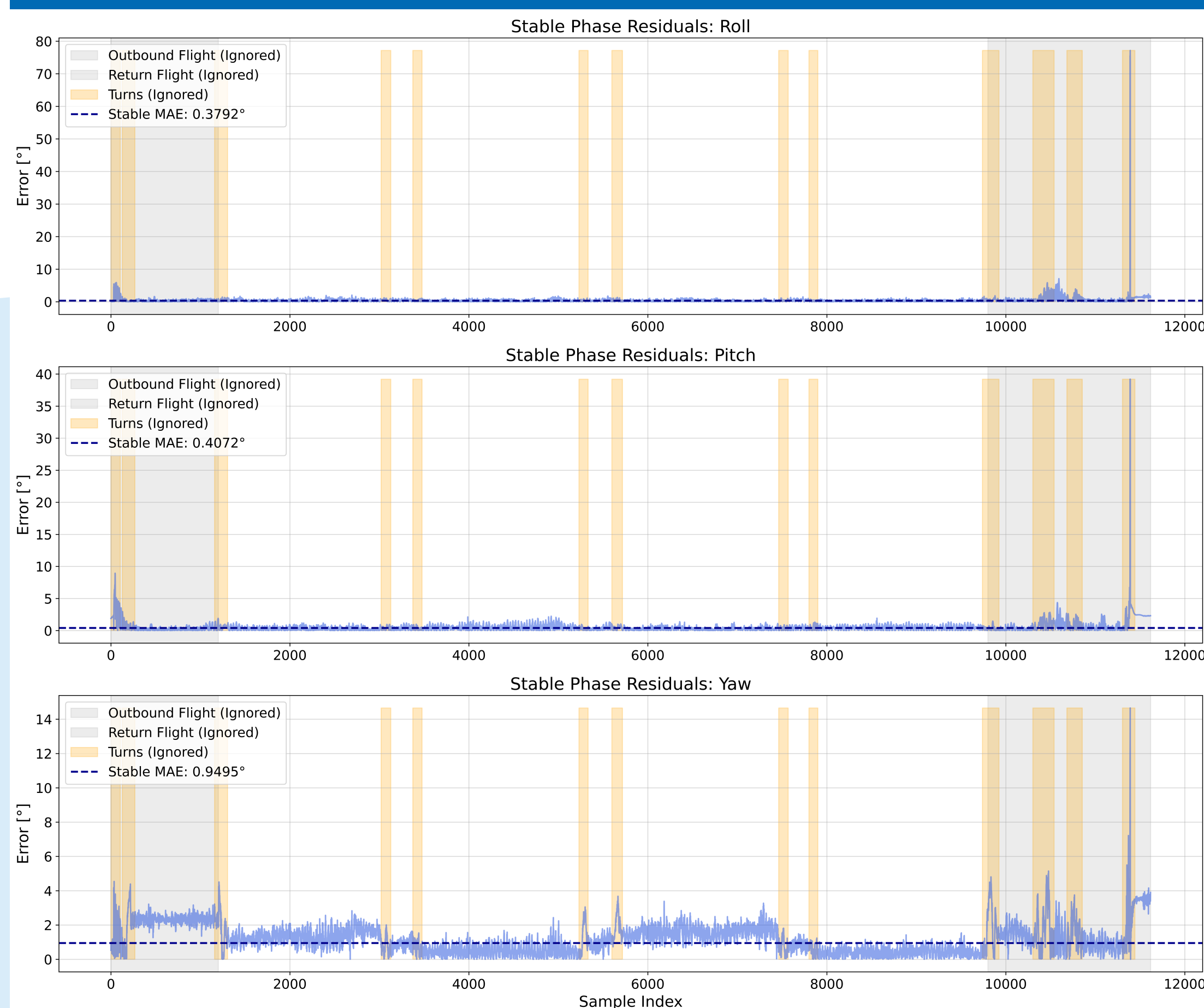


Fig. 6: Distribution of the model's absolute prediction errors for Roll, Pitch, and Yaw.

Conclusion

Conclusion: The Transformer architecture demonstrates exceptional performance in predicting time-dependent orientation angles, significantly outperforming the FFN baseline [2]. While the FFN struggles to capture high-frequency oscillations, the Transformer tracks the ground truth with high fidelity across all three axes. Notably, the prediction of the yaw angle—previously a challenge—now shows high stability with a Mean Absolute Error (MAE) below 1°. Residual analysis confirms that the remaining errors are primarily confined to the start and end of the sequence, where the limited temporal context inherently constrains the model's predictive power.

Outlook: Future work will investigate methods to mitigate edge-effect errors, for instance, through advanced padding techniques or bidirectional processing. Additionally, exploring state-space models like Mamba or hybrid Mamba-Transformer architectures remains a promising avenue to further enhance computational efficiency while maintaining this high level of precision.

FFN Prediction

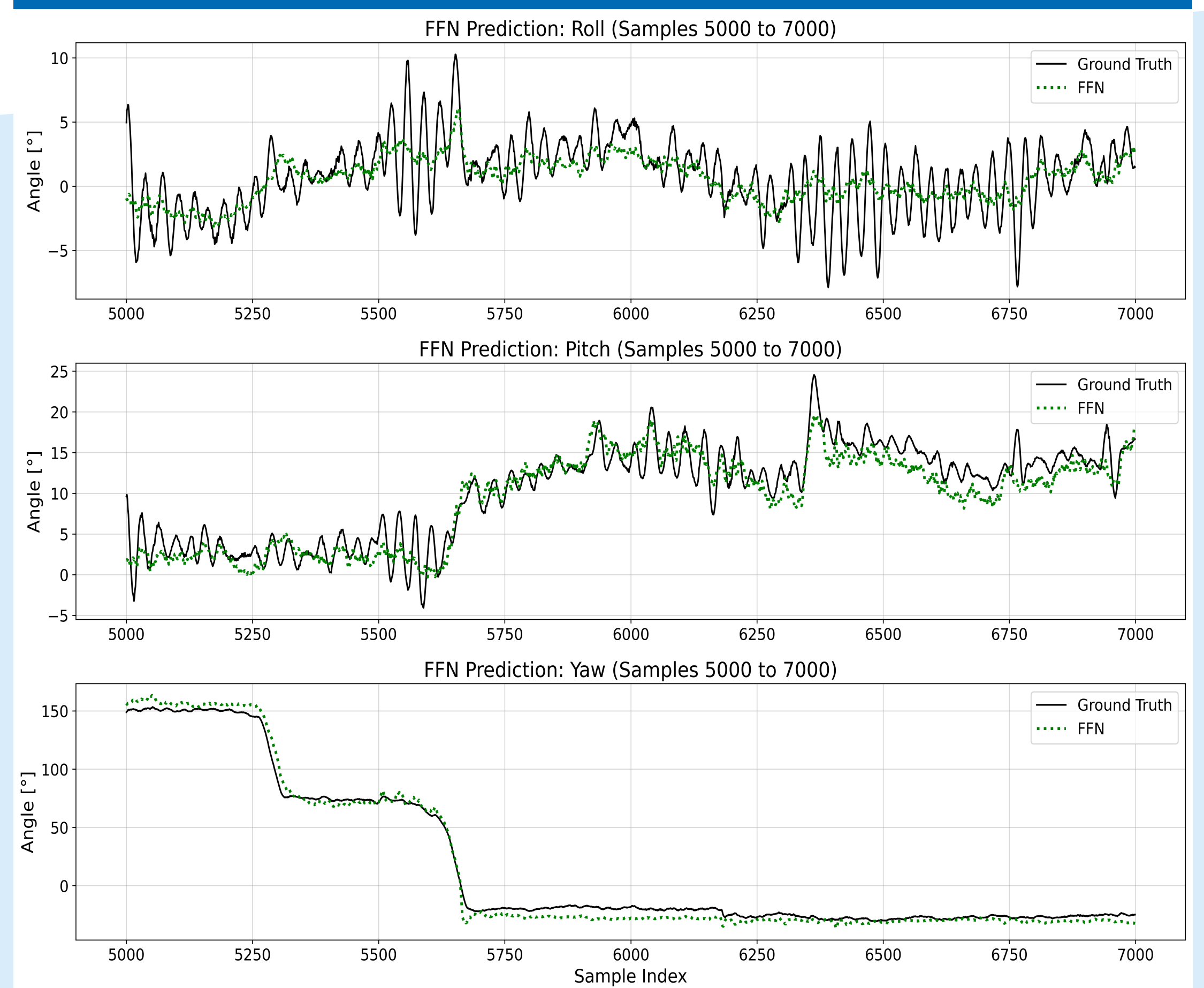


Fig. 5: Prediction results for the FFN model [2], showing its performance on the same test segment as the Transformer.

References

- [1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. arXiv. <https://doi.org/10.48550/arXiv.1706.03762>
- [2] Kotowski, P. O., Becken, M., 2024. Forecasting Drone-towed Sensor Platform Orientation with Neural Networks. 84. Jahrestagung der Deutschen Geophysikalischen Gesellschaft (DGG), Jena.